

Derivation of a Quantile Confidence Interval Based on Order Statistics

Frederik Danielsen

This note provides a mathematical derivation of a confidence interval for a quantile ξ_q based on order statistics. The results presented here are classical and are not original to this note. They are derived independently to ensure a thorough understanding of the underlying mathematics, to present the arguments in a form suited to the Problab¹ project, and to obtain conclusions that are directly useful for the implementation in Problab. The derivation assumes i.i.d. samples $X_1, \dots, X_n$ satisfying $P(X_i \leq \xi_q) = q$ and a significance level $\alpha \in (0, 1)$, corresponding to a target coverage level of $1 - \alpha$. In the first lemma, a confidence interval of the form $[X_{(k_L^*)}, X_{(k_U^*+1)}]$ is derived by choosing the indices k_L^* and k_U^* such that the two binomial tail probabilities are bounded by $\alpha/2$. In the second lemma, it is shown that the iterative procedures used in the implementation terminate at exactly these optimal indices.

Practically, these results enable Problab to return finite-sample confidence intervals with coverage probability at least $1 - \alpha$ for estimated quantiles, such as medians or percentiles.

Lemma 1. *Let X_i with $i \in \{1, \dots, n-1\}$ be i.i.d samples. Suppose that $P(X_i \leq \xi_q) = q$ for all i , some $q \in (0, 1)$, and ξ_q in the value space of X_i . Let*

$$I_i = \begin{cases} 1, & X_i \leq \xi_q, \\ 0, & X_i > \xi_q. \end{cases}$$

and let

$$K = \sum_{i=1}^n I_i \sim \text{Bin}(n, q).$$

Let k_L^* and k_U^* solve

$$\begin{aligned} k_L^* &= \max_{k_L \in \{1, \dots, n-1\}} \left\{ k_L : P(K < k_L) \leq \frac{\alpha}{2} \right\} \\ k_U^* &= \min_{k_U \in \{1, \dots, n-1\}} \left\{ k_U : P(K > k_U) \leq \frac{\alpha}{2} \right\}. \end{aligned}$$

Then

$$[X_{(k_L^*)}, X_{(k_U^*+1)}]$$

is a confidence interval for ξ_q with coverage probability at least $1 - \alpha$. Furthermore, among the intervals satisfying the two tail constraints, solving the optimization problems above produces the narrowest interval.

¹<https://github.com/FrederikDanielsenDevelopment/problab>

Proof. I_i can be regarded as the outcomes of individual Bernouli trials with probability of success q with

$$K = \sum_{i=1}^n I_i \sim \text{Bin}(n, q)$$

denoting the number of samples with values at most ξ_q . Now, consider the ordered samples

$$X_{(1)}, X_{(2)}, \dots, X_{(n)}.$$

If $K \leq k_U$ for some $k_U \in \{1, \dots, n-1\}$, that would imply that at most k_U samples fell below ξ_q . That is, $\xi_q < X_{(k_U+1)}$. Likewise, if $K \geq k_L$ for some $k_L \in \{1, \dots, n-1\}$, that would imply that at least k_L samples fell below ξ_q . That is, $\xi_q \geq X_{(k_L)}$. Hence,

$$P(k_L \leq K \leq k_U) = P(X_{(k_L)} \leq \xi_q < X_{(k_U+1)}).$$

Therefore, if we determine k_L and k_U such that

$$P(k_L \leq K \leq k_U) = 1 - \alpha,$$

for some $\alpha \in (0, 1)$, then the interval $[X_{(k_L)}, X_{(k_U+1)}]$ is a confidence interval for ξ_q with coverage probability $1 - \alpha$. However, this equality is one with two unknowns and thus multiple solutions. However, we may impose an additional constraint, namely that the likelihood that ξ_q lies above the confidence interval is the same as the likelihood that it lies below it – at least to the extent that a discrete number of samples allows it. Specifically, we require that

$$P(\xi_q \geq X_{(k_U+1)}) = P(\xi_q < X_{(k_L)}) = \alpha/2.$$

However, as mentioned, since we are dealing with a discrete number of samples, it is likely that no (k_L, k_U) exists that fulfil this exactly. Therefore, we make do with the best estimate (likely more conservative than $1 - \alpha$) and require simply that

$$P(\xi_q \geq X_{(k_U+1)}) \leq \alpha/2,$$

$$P(\xi_q < X_{(k_L)}) \leq \alpha/2$$

$$\Leftrightarrow$$

$$P(K > k_U) \leq \alpha/2,$$

$$P(K < k_L) \leq \alpha/2.$$

Since $\alpha < 1$, any pair (k_L, k_U) satisfying both tail constraints necessarily satisfies $k_L \leq k_U$. Indeed, if $k_L > k_U$, then every possible value of K satisfies either $K < k_L$ or $K > k_U$, and therefore

$$1 = P((K < k_L) \cup (K > k_U)) \leq P(K < k_L) + P(K > k_U) \leq \alpha,$$

which is a contradiction. Furthermore, in case you were wondering, since the lower endpoint $X_{(k_L)}$ requires $k_L \geq 1$ and the upper endpoint $X_{(k_U+1)}$ requires $k_U \leq n-1$, the inequality $k_L \leq k_U$ implies that any feasible pair must satisfy

$$k_L, k_U \in \{1, \dots, n-1\}.$$

To obtain the narrowest confidence interval satisfying these constraints, we choose the largest possible k_L and the smallest possible k_U . Thus, we solve

$$k_L^* = \max_{k_L \in \{1, \dots, n-1\}} \left\{ k_L : P(K < k_L) \leq \frac{\alpha}{2} \right\},$$

$$k_U^* = \min_{k_U \in \{1, \dots, n-1\}} \left\{ k_U : P(K > k_U) \leq \frac{\alpha}{2} \right\}.$$

In this way we ensure that

$$P(k_L^* \leq K \leq k_U^*) = P(X_{(k_L^*)} \leq \xi_q < X_{(k_U^*+1)}) \geq 1 - \alpha.$$

Then the interval

$$[X_{(k_L^*)}, X_{(k_U^*+1)}]$$

is at least as conservative as a confidence interval with coverage probability $1 - \alpha$. $\square$

Lemma 2. *Let $\alpha \in (0, 1)$, $q \in (0, 1)$, $n \geq \lceil \ln(\alpha/2) / \ln(\max(q, 1 - q)) \rceil$ and $K \sim \text{Bin}(n, q)$. Then the optimization problems*

$$k_L^* = \max_{k \in \{1, \dots, n-1\}} \left\{ k : P(K < k) \leq \frac{\alpha}{2} \right\},$$

$$k_U^* = \min_{k \in \{1, \dots, n-1\}} \left\{ k : P(K > k) \leq \frac{\alpha}{2} \right\}.$$

have solutions.

Proof. Suppose that no feasible pair (k_L^*, k_U^*) exists. Since $P(K < k)$ is increasing and $P(K > k)$ is decreasing for $k \in \{1, \dots, n-1\}$, that would imply that

$$P(K < 1) = P(K = 0) = (1-q)^n > \alpha/2 \bigvee P(K > n-1) = P(K = n) = q^n > \alpha/2$$

$$\Leftrightarrow \max(q^n, (1-q)^n) = \max(q, 1-q)^n > \alpha/2.$$

Therefore, if we want

$$P(K < 1) \leq \alpha/2 \wedge P(K > n-1) \leq \alpha/2,$$

then it must be the case that

$$\max(q, 1-q)^n \leq \frac{\alpha}{2}.$$

Taking logarithms gives

$$n \ln(\max(q, 1-q)) \leq \ln\left(\frac{\alpha}{2}\right).$$

Since $\max(q, 1-q) \in (0, 1)$, we have

$$\ln(\max(q, 1-q)) < 0,$$

so dividing by $\ln(\max(q, 1 - q))$ reverses the inequality:

$$n \geq \frac{\ln(\alpha/2)}{\ln(\max(q, 1 - q))}.$$

Thus, the minimum value required for n is

$$n_{\min} = \left\lceil \frac{\ln(\alpha/2)}{\ln(\max(q, 1 - q))} \right\rceil.$$

Hence, under the stated condition on n , both optimization problems have solutions. $\square$

Lemma 3. *Let $K \sim \text{Bin}(n, q)$ and $\alpha \in (0, 1)$, and suppose that the optimization problems*

$$k_L^* = \max_{k \in \{1, \dots, n-1\}} \left\{ k : P(K < k) \leq \frac{\alpha}{2} \right\},$$

$$k_U^* = \min_{k \in \{1, \dots, n-1\}} \left\{ k : P(K > k) \leq \frac{\alpha}{2} \right\}$$

have solutions. Consider the iterative procedures

$$k_L \leftarrow 1,$$

$$\text{while } k_L < n - 1 \text{ and } P(K \leq k_L) \leq \frac{\alpha}{2}, \quad k_L \leftarrow k_L + 1,$$

and

$$k_U \leftarrow n - 1,$$

$$\text{while } k_U > 1 \text{ and } P(K \geq k_U) \leq \frac{\alpha}{2}, \quad k_U \leftarrow k_U - 1.$$

Then the procedures terminate with

$$k_L = k_L^* \quad \text{and} \quad k_U = k_U^*.$$

Proof. For the lower index, observe that increasing the current value k_L to $k_L + 1$ is permissible exactly when

$$P(K < k_L + 1) = P(K \leq k_L) \leq \frac{\alpha}{2}.$$

Thus, at every iteration, the algorithm increments k_L only when the next larger value satisfies the lower-tail constraint. Since k_L increases by one at each iteration and is bounded above by $n - 1$, the procedure terminates. If the procedure stops before reaching $n - 1$, then increasing k_L by one would violate the constraint. Indeed, the next candidate would be $k_L + 1$, and for this candidate the constraint is

$$P(K < k_L + 1) \leq \frac{\alpha}{2}.$$

Since

$$P(K < k_L + 1) = P(K \leq k_L),$$

and the loop has stopped, this inequality is no longer satisfied. Therefore, $k_L + 1$ is not feasible. Since $P(K < k)$ is nondecreasing as k increases, no larger value can be feasible either. Hence, the returned k_L is the largest feasible value, so

$$k_L = \max_{k \in \{1, \dots, n-1\}} \left\{ k : P(K < k) \leq \frac{\alpha}{2} \right\} = k_L^*.$$

If the procedure instead terminates at $k_L = n - 1$, then $n - 1$ is itself the largest possible value in the domain, so again $k_L = k_L^*$. For the upper index, observe that decreasing the current value k_U to $k_U - 1$ is permissible exactly when

$$P(K > k_U - 1) = P(K \geq k_U) \leq \frac{\alpha}{2}.$$

Thus, at every iteration, the algorithm decrements k_U only when the next smaller value satisfies the upper-tail constraint. Since k_U decreases by one at each iteration and is bounded below by 1, the procedure terminates. If the procedure stops before reaching 1, then decreasing k_U by one would violate the constraint. Indeed, the next candidate would be $k_U - 1$, and for this candidate the constraint is

$$P(K > k_U - 1) \leq \frac{\alpha}{2}.$$

Since

$$P(K > k_U - 1) = P(K \geq k_U),$$

and the loop has stopped, this inequality is no longer satisfied. Therefore, $k_U - 1$ is not feasible. Since $P(K > k)$ increases as k decreases, no smaller value can be feasible either. Hence, the returned k_U is the smallest feasible value, so

$$k_U = \min_{k \in \{1, \dots, n-1\}} \left\{ k : P(K > k) \leq \frac{\alpha}{2} \right\} = k_U^*.$$

If the procedure instead terminates at $k_U = 1$, then 1 is itself the smallest possible value in the domain, so again $k_U = k_U^*$. $\square$